Project Report: Book Management System

The Book Management System is the name of the software product that is aimed to provide users with various tools as well as features for keeping their libraries available, organized, and updated. It is the brainchild of the 'Vugar-Laman-Musa-Huseyn' team comprising of book lovers who have created this system to make it as user friendly as possible. This means that nobody will find it difficult to organize their reading resources. We shall have a look at the components, design, bringing to realities of the Book Management System.

Overview:

The Book Management System allows users to:

- Explore a database of books.
- Filter and sort books based on various criteria.
- View book details such as title, author, rating, and reviews.
- Add selected books to their personal library.
- Control their book collection efficiently.

Features:

1. Book Exploration: The system comes with the users having the opportunity to browse a catalogue of books. Through that, they are able to visit numerous authors and titles. The user interface has been created in a way that offers intuitiveness at its best with a clean layout that allows the user to access the books he/she wants quickly and easily.
2. Filtering and Sorting: Readers can choose books by title or author of the books during the search. In addition, they will be able to sort by book title, author, rating, and reviews, which are search features not offered to visitors. These functionalities make it easy for readers to find the titles of interest among lasting screen.
3. Book Details: Users can check all attributes of each title including its name, author, rating, and also small description (reviews). It empowers them to get acquainted with the publishing industry and eventually make their own decisions about the books that should be on their library list.
4. Adding to Library: By simply picking books of their interest, users can add them to their collection. The system checks for duplicates entries so that one cannot be recorded twice. Therefore, in this way, the system maintains the user's collection overall quality.
5. User Authentication: The newly added user authentication feature requires users to log in before accessing the system. This enhances security and allows for personalized user experiences.
6. Login and Registration: People, who are interested in using the system, are required to create an account and get access to that system. When people enroll, they have to come up with a username and a password that they'll be using to enter the system later.

Contributions:

Full name	Student ID	Approximate % of work done	What was Done
Laman Jananova	18034	30%	Implemented basic interface, including user login, registration and basics of personal and general database, as well synchronized and visible user reviews and average ratings for books
Huseyn Mirzayev	000019697	30%	Implemented clickable and editable functionalities, improved general database author viewing, worked on personal database management, helped with admin part
Musa Aliyev		20%	Helped with the implementation of personal and general database and worked on admin part
Vugar Mirzayev		20%	Worked on sorting and filtering all the books in general and personal databases as well as the search functionality.

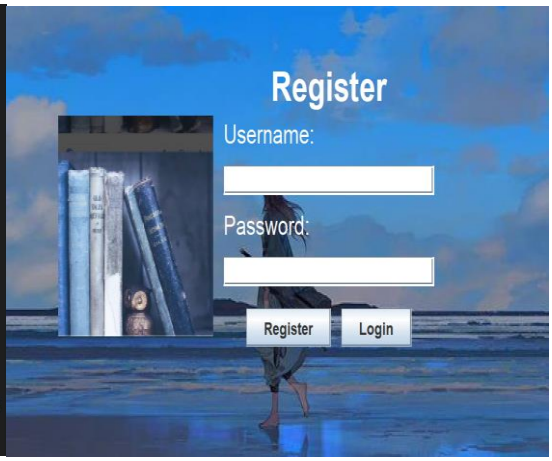
Conclusion:

The Book Management System designed by Vugar, Laman, Musa, and Huseyn is a total package for the purpose of book keeping as well as for the administration of the

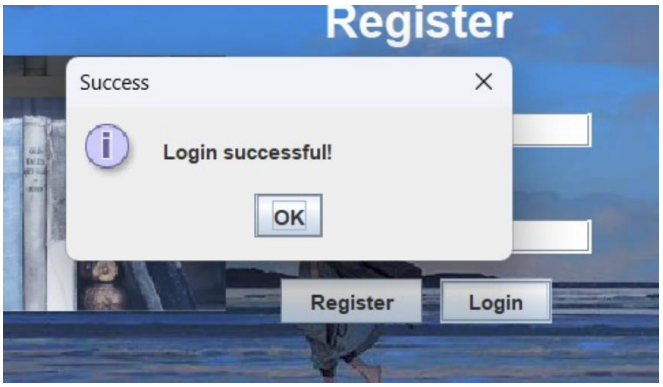
books collection. The system does that with such an intuitive interface, the best features and the modular design so, that the users can explore, to filter and add books to their library in a smooth manner. From now on, investments in user authentication, book recommendations, and feedback systems from users will be done to expand the capabilities and simplicity of the system.

1)MyGui.java

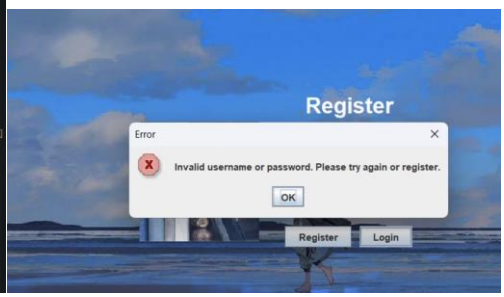
```java
public class MyGUI extends JFrame {
    private static final String DATABASE_FILE = "users.csv";
    private Map<String, User> usersMap; // Map to store usernames and User objects

    public MyGUI() {
        setTitle(title:"Registration Form");
        // setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setResizable(resizable:false);
        ImageIcon icon = new ImageIcon(filename:"Images/left.jpeg");
        setIconImage(icon.getImage());

        // Set background image
        ImageIcon backgroundImage = new ImageIcon(filename:"Images/back.jpeg");
        JLabel backgroundLabel = new JLabel(backgroundImage);
        setContentPane(backgroundLabel);
        setLayout(new BorderLayout());

        // Load existing users from CSV database into memory
        usersMap = loadUsersFromCSV();

        // Create a panel for the left image and registration
        JPanel centerPanel = new JPanel(new GridBagLayout());
        centerPanel.setOpaque(isOpaque:false); // Make panel transparent
        add(centerPanel, BorderLayout.CENTER);

        // Left side (Picture)
        ImageIcon leftImageIcon = new ImageIcon(filename:"Images/left.jpeg");
        Image scaledImage = leftImageIcon.getImage().getScaledInstance(width:150, height:150, Image.SCALE_SMOOTH);
        leftImageIcon = new ImageIcon(scaledImage);
        JLabel leftImageLabel = new JLabel(leftImageIcon);
```




```java
        // Validate login credentials for regular users
        User user = usersMap.get(username);
        if (user != null && user.getPassword().equals(password)) {
            JOptionPane.showMessageDialog(this, message:"Login successful!", title:"Success",
                    JOptionPane.INFORMATION_MESSAGE);
            new ShowBooks(username).setVisible(b:true); // Open the regular user interface
            dispose(); // Close the current login window
        } else {
            JOptionPane.showMessageDialog(this, message:"Invalid username or password. Please try again or register.",
                    title:"Error", JOptionPane.ERROR_MESSAGE);
        }
    });


    // Add buttons to buttonPanel
    buttonPanel.add(registerButton);
    buttonPanel.add(loginButton);

    // Add formPanel and buttonPanel to registrationPanel
    registrationPanel.add(formPanel, BorderLayout.CENTER);
    registrationPanel.add(buttonPanel, BorderLayout.SOUTH);

    // Add leftImageLabel and registrationPanel to centerPanel
    centerPanel.add(leftImageLabel);
    centerPanel.add(registrationPanel);

    pack();
    setLocationRelativeTo(c:null); // Center the frame on the screen
```



```java
// Register and Login Buttons
JPanel buttonPanel = new JPanel(new FlowLayout(FlowLayout.CENTER, hgap:10, vgap:10));
buttonPanel.setOpaque(isOpaque:false); // Make panel transparent
JButton registerButton = new JButton(text:"Register");
JButton loginButton = new JButton(text:"Login");

// Action listener for Register button
registerButton.addActionListener(e -> {
    String username = usernameField.getText();
    String password = new String(passwordField.getPassword());

    // Validate input fields
    if (username.isEmpty() || password.isEmpty()) {
        JOptionPane.showMessageDialog(this, message:"Please enter both username and password.", title:"Error",
                JOptionPane.ERROR_MESSAGE);
        return;
    }

    // Check password strength
    if (!isPasswordStrong(password)) {
        JOptionPane.showMessageDialog(this,
                message:"Password must contain at least one uppercase letter, one lowercase letter, and be at least 8 characters l
                title:"Error", JOptionPane.ERROR_MESSAGE);
        return;
    }

    // Check if username already exists
    if (usersMap.containsKey(username)) {
        JOptionPane.showMessageDialog(this, message:"Username already exists. Please choose a different username.",
                title:"Error", JOptionPane.ERROR_MESSAGE);
    } else {
```



2)General database -displaying books to user

```java
public ShowBooks(String username) {
    createCsvCopy();  // Create a copy of the CSV file at initialization
    setTitle(title:"Explore Books");
    setSize(width:800, height:600);
    setLocationRelativeTo(c:null);
    setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

    // Header panel with BorderLayout
    JPanel headerPanel = new JPanel(new BorderLayout());

    // Header label
    JLabel headerLabel = new JLabel(text:"Books Library");
    headerLabel.setFont(new Font(name:"Monospaced", Font.PLAIN, size:33));
    headerLabel.setForeground(Color.BLACK);
    headerLabel.setHorizontalAlignment(SwingConstants.CENTER);
    headerPanel.add(headerLabel, BorderLayout.CENTER);

    // Back button panel with FlowLayout
    JPanel backButtonPanel = new JPanel(new FlowLayout(FlowLayout.LEFT));

    // Back button
    JButton backButton = new JButton(text:"Back");
    backButton.addActionListener(e -> {
        MyGUI loginPage = new MyGUI();
        loginPage.setVisible(b:true);
      dispose();
      dispose();
    });
    backButton.setBackground(Color.WHITE);
    backButtonPanel.add(backButton);
```

```java
// Search field
searchField = new JTextField(columns:20);
searchField.setToolTipText(text:"Enter title or author to search");
searchPanel.add(searchField);

// Search button
JButton searchButton = new JButton(text:"Search");
searchButton.addActionListener(e -> {
    String query = searchField.getText().trim().toLowerCase();
    filterBooks(query);
});

searchPanel.add(searchButton);

// Add search panel to NORTH of the frame
headerPanel.add(searchPanel, BorderLayout.EAST);

// Initialize table model
tableModel = new DefaultTableModel() {
    @Override
    public Class<?> getColumnClass(int columnIndex) {
        if (columnIndex == 4) {
            return Boolean.class;
        } else {
            return String.class;
        }
    }

    @Override
    public boolean isCellEditable(int row, int column) {
        return column == 4;
    }
};
tableModel.addColumn(columnName:"Title");
tableModel.addColumn(columnName:"Author");
tableModel.addColumn(columnName:"Rating");
tableModel.addColumn(columnName:"Reviews");
tableModel.addColumn(columnName:"Selected");

// Read data from CSV and populate the table
readDataFromCSV();
```

Title	Author	Rating	Reviews	Selected
Mahabharata [Bhagavad Git...	Unknown	3.00 (2)	laman, Vugar	
Epic of Gilgamesh	Unknown	3.50 (2)	laman, huseyn	
The Old Testament	Unknown	4.00 (2)	Vugar, huseyn	
Iliad	Homer	5.00 (1)	laman	
Odyssey	Homer	4.00 (1)	laman	
Histories	Herodotus	No Rating	No reviews	
[Plays]	Sophocles	3.00 (1)	laman	
[Plays]	Aeschyles	3.00 (1)	laman	
Hippolytus	Euripides	No Rating	No reviews	
Bachants	Euripides	5.00 (1)	laman	
Electra	Euripides	1.00 (1)	Vugar	
The Phornician Women	Euripides	2.50 (2)	laman, Vugar	
The Peloponesian War	Thucydides	5.00 (1)	Vugar	
Dialogues	Plato	No Rating	No reviews	
Poetics	Aristotle	No Rating	No reviews	

3)personal library

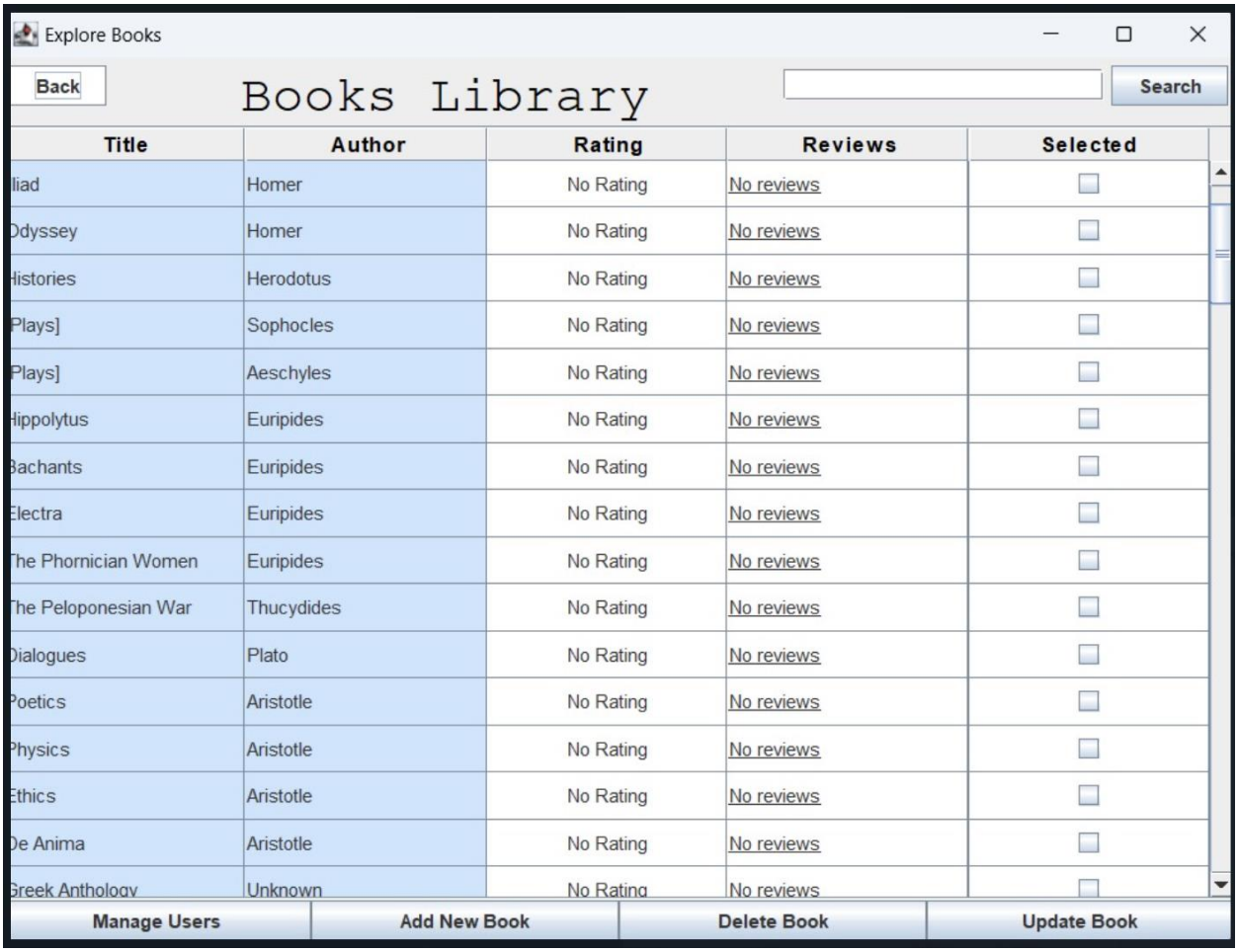
```java
setTitle(title:"Personal Library");
setSize(width:900, height:400);
setLocationRelativeTo(c:null);

// Set up username label at the top
JLabel usernameLabel = new JLabel("Personal Library of " + username);
usernameLabel.setFont(new Font(name:"Monospaced", Font.PLAIN, size:25));
usernameLabel.setHorizontalAlignment(SwingConstants.CENTER);

// Create a panel for the username label and the back button
JPanel topPanel = new JPanel(new BorderLayout());
topPanel.add(usernameLabel, BorderLayout.CENTER);

// Back button
JButton backButton = new JButton(text:"Back");
backButton.addActionListener(e -> {
    dispose(); // Close the current frame
    ShowBooks showBooks = new ShowBooks(username); // Create new instance of ShowBooks
    showBooks.setVisible(b:true); // Show the ShowBooks frame
});
backButton.setBackground(Color.WHITE);

// Add the back button to the panel
topPanel.add(backButton, BorderLayout.WEST);

// Add the panel to the NORTH of the content pane
getContentPane().add(topPanel, BorderLayout.NORTH);
```

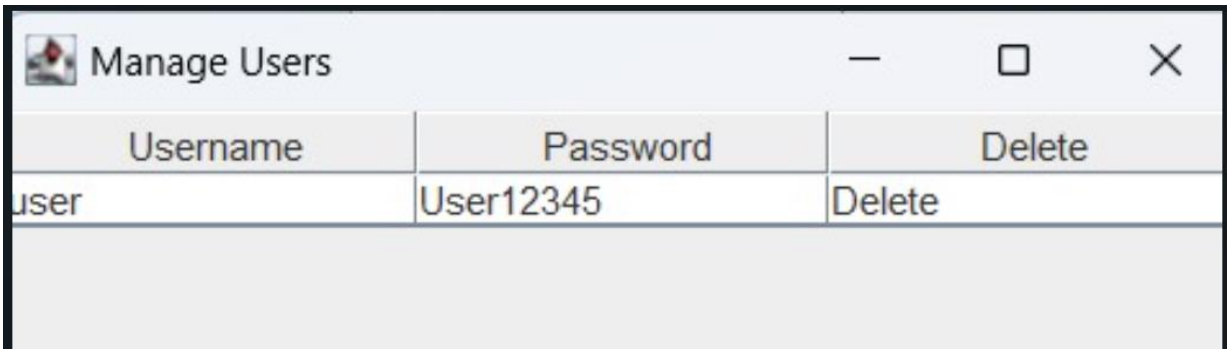
```java
private void showUserDetailsDialog(String title, String author, String averageRating, String userRating, String userReview, int rowIndex) {
    JFrame userDetailsFrame = new JFrame(title:"User Details");
    userDetailsFrame.setSize(width:400, height:300);
    userDetailsFrame.setLocationRelativeTo(c:null);
    userDetailsFrame.setDefaultCloseOperation(JFrame.DISPOSE_ON_CLOSE);

    JPanel userDetailsPanel = new JPanel();
    userDetailsPanel.setLayout(new BoxLayout(userDetailsPanel, BoxLayout.Y_AXIS));
    userDetailsPanel.setBorder(BorderFactory.createEmptyBorder(top:10, left:10, bottom:10, right:10));

    JLabel titleLabel = new JLabel("Title: " + title);
    JLabel authorLabel = new JLabel("Author: " + author);
    JLabel ratingLabel = new JLabel("Average Rating: " + averageRating);
    JLabel userRatingLabel = new JLabel(text:"User Rating:");

    // Create a combo box for user rating with options 1 through 5
    String[] ratingOptions = {"1", "2", "3", "4", "5"};
    JComboBox<String> userRatingComboBox = new JComboBox<>(ratingOptions);
    userRatingComboBox.setSelectedItem(userRating);

    JLabel userReviewLabel = new JLabel(text:"User Review:");
    JTextArea userReviewArea = new JTextArea(userReview, rows:8, columns:30);

    // Add a vertical scroll pane to the user review text area
    JScrollPane scrollPane = new JScrollPane(userReviewArea);
    scrollPane.setVerticalScrollBarPolicy(ScrollPaneConstants.VERTICAL_SCROLLBAR_ALWAYS);

    userDetailsPanel.add(titleLabel);
    userDetailsPanel.add(authorLabel);
    userDetailsPanel.add(ratingLabel);
    userDetailsPanel.add(userRatingLabel);
    userDetailsPanel.add(userRatingComboBox);
    userDetailsPanel.add(userReviewLabel);
    userDetailsPanel.add(scrollPane);
```

4) admin page

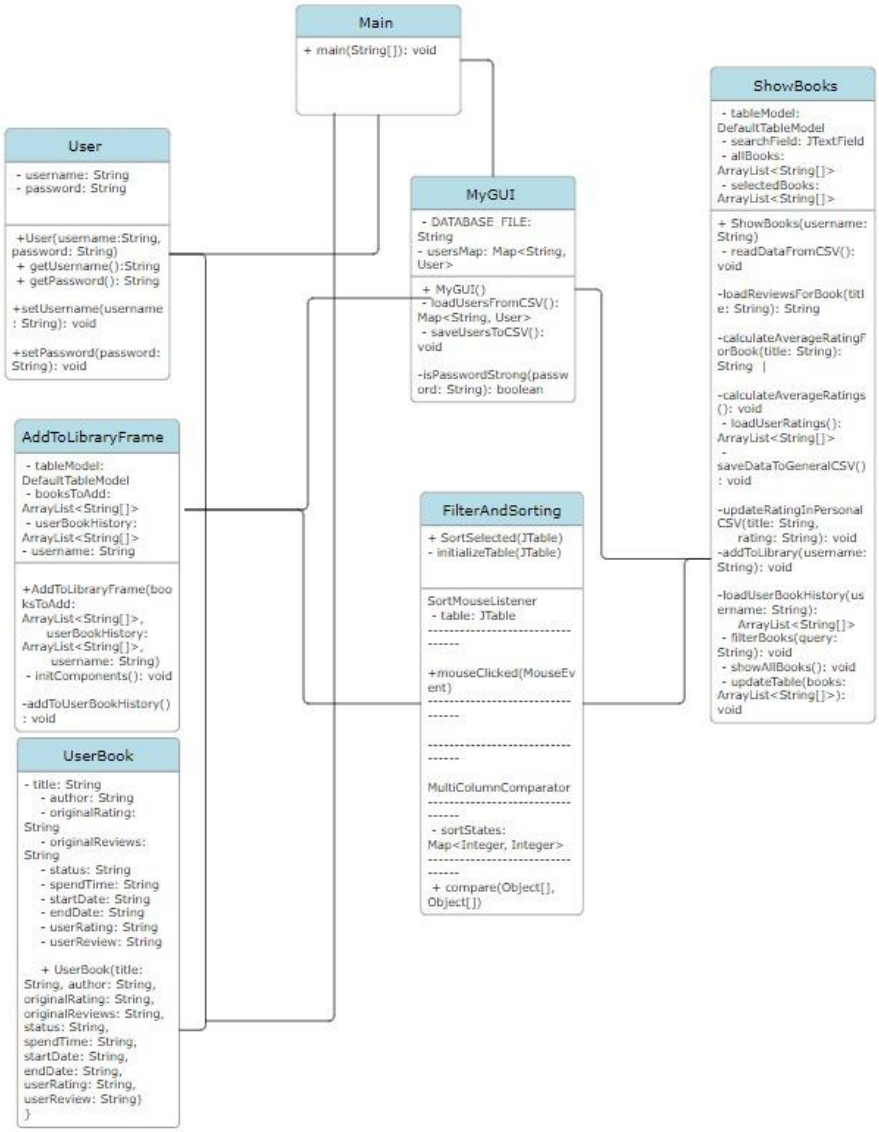
```java
JPanel adminPanel = new JPanel(new GridLayout(rows:1, cols:4));

JButton deleteUserButton = new JButton(text:"Manage Users");
deleteUserButton.addActionListener(e -> showUserManagementDialog());

JButton addBookButton = new JButton(text:"Add New Book");
addBookButton.addActionListener(e -> addNewBook());

JButton deleteBookButton = new JButton(text:"Delete Book");
deleteBookButton.addActionListener(e -> deleteBook());

JButton updateBookButton = new JButton(text:"Update Book");
updateBookButton.addActionListener(e -> updateBook());

adminPanel.add(deleteUserButton);
adminPanel.add(addBookButton);
adminPanel.add(deleteBookButton);
adminPanel.add(updateBookButton);

add(adminPanel, BorderLayout.SOUTH);

FilterAndSorting.SortSelected(bookTable);
setVisible(b:true);

calculateAverageRatings();

saveDataToGeneralCSV();
```

```java
private void showUserManagementDialog() {
    JFrame userManagementFrame = new JFrame(title:"Manage Users");
    userManagementFrame.setSize(width:400, height:300);
    userManagementFrame.setLocationRelativeTo(this);

    // Boolean flag to track if any user was deleted
    final boolean[] userDeleted = new boolean[1];

    ArrayList<String[]> userData = readUserData(filename:"users.csv");

    DefaultTableModel userModel = new DefaultTableModel();
    userModel.addColumn(columnName:"Username");
    userModel.addColumn(columnName:"Password");
    userModel.addColumn(columnName:"Delete");

    for (String[] userDataEntry : userData) {
        Object[] row = { userDataEntry[0], userDataEntry[1], "Delete" };
        userModel.addRow(row);
    }

    JTable userTable = new JTable(userModel);
    userTable.addMouseListener(new MouseAdapter() {
        @Override
        public void mouseClicked(MouseEvent e) {
            int column = userTable.getColumnModel().getColumnIndex(columnIdentifier:"Delete");
            int row = userTable.rowAtPoint(e.getPoint());

            if (column == 2 && row != -1) { // Clicked on delete button
                String usernameToDelete = (String) userModel.getValueAt(row, column:0);
                deleteUserData(usernameToDelete);
                userModel.removeRow(row);
                userDeleted[0] = true; // Set flag to true as a user has been deleted
            }
        }
    });

    JScrollPane scrollPane = new JScrollPane(userTable);
    userManagementFrame.add(scrollPane);
    userManagementFrame.setVisible(b:true);
```





Github link -https://github.com/ADA-SITE-CSCI-1202/team-project-team-25.git

Video link -https://www.youtube.com/watch?v=T2Xv536vMPk

Class diagram